



INDIVIDUAL DEFENSE KIT

Vital Spring Medical Center Defense Cheat Sheet

Care in motion. Information in place.

INS 204 · System Analysis and Design
Group 9 · 200 Level

Use this sheet to recall the structure and reasoning. Do not read it word-for-word during the defense.

1. The Project in 60 Seconds

Opening answer

Vital Spring Medical Center is treated as a medium-sized Nigerian outpatient facility that currently relies on paper folders, handwritten registers and a physical appointment book. The proposed Clinic Appointment System replaces the fragmented process with a secure browser-based service for patient registration, practitioner availability, appointment booking, patient check-in, live queue coordination, visit documentation, reminders and management reporting.

Core problem • Paper records are disconnected and easily lost. • Double-booked slots and missing files. • No automatic reminders or dependable statistics. • No accountable access to sensitive information.	Core value • One source of truth for patients and slots. • Live queue tracking reduces waiting times. • Automated confirmations and reminders. • Measurable attendance, wait and utilisation.
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2. Numbers You Must Know

18 + 12 functional + non-functional requirements	6 + 6 Level-0 processes + data stores
10 principal database entities	22 weeks proposed implementation period
₦7.30m one-time investment	₦1.40m annual recurring cost
₦4.50m annual quantified benefits	2.35 years payback; five-year BCR ≈ 3.2

Economic figures are illustrative 2026 classroom estimates, not vendor quotations.

3. Objectives and Boundaries

Success targets	Deliberately out of scope
Zero double-booking; patient search under 5 seconds; existing-patient booking under 2 minutes; median wait reduced 20%; 99% clinic-hours availability; quarterly restoration within 4 hours.	Billing and payments, insurance claims, pharmacy stock, laboratory integration, telemedicine, external health-record exchange and automated diagnosis.

4. Models: What Each Diagram Proves

Model	What it shows	Defense sentence
Context DFD	One system process and five external entities	It defines the system boundary; internal stores are intentionally omitted because the context diagram abstracts implementation detail.
Level-0 DFD	Six internal processes and six stores	It decomposes Vital Spring; Process 3.0 Coordinate Appointments is central because it validates slots, reserves one, records the booking source and triggers communication.
ERD	Entities, attributes, PK/FK, cardinality and optionality	It defines the logical data structure, not data movement. Queue entries and availability slots are separate entities because an appointment reserves a planned slot while the queue captures actual arrival.
Architecture	Four access channels, web gateway, five application services	Clients never access the database directly; all requests pass through the secure web gateway and a dedicated application service.

Six Level-0 processes

1. Register and Identify Patients
2. Maintain Practitioner Schedules
3. Coordinate Appointments
4. Manage Arrival and Queue
5. Document Patient Visit
6. Communicate and Report

Ten entities

1. Patient
2. Department
3. Practitioner
4. Availability Slot
5. Appointment
6. Queue Entry
7. Visit Record
8. Notification
9. User Account
10. Activity Log

5. ERD and Normalization

Key cardinalities

- Patient 1:M Appointment Practitioner 1:M Appointment Department 1:M Practitioner
- Appointment 1:1 Availability Slot Appointment 1:0..1 Visit Record Appointment 1:M Queue Entry
- User Account 1:M Activity Log

Stage	Change	Anomaly removed
UNF	Paper register contains repeating appointment groups inside a single patient row.	Starting problem
1NF	Atomic values; one appointment per row.	Repeating groups
2NF	Patient facts separated from composite appointment data.	Partial dependencies
3NF	Practitioner, department and location facts separated.	Transitive dependencies

Three anomalies

Update: changing Dr. Ibekwe's department everywhere. **Insertion:** unable to add a practitioner without an appointment. **Deletion:** deleting the only appointment removes patient facts.

6. Strongest Technical Defense

How double-booking is prevented

1. The interface displays only open, future slots.
2. The server locks the selected slot, rechecks its status and reserves it inside one database transaction.
3. A unique constraint (unique slot_id on active appointment) permits only one appointment per slot.
4. If two users confirm simultaneously, one commit succeeds; the other rolls back and receives available alternatives.

```
BEGIN;  
SELECT status FROM availability_slot  
WHERE id = :slot_id FOR UPDATE;  
-- application checks status is 'open'  
INSERT INTO appointment (slot_id, patient_id, ...)  
VALUES (:slot_id, :patient_id, ...);  
UPDATE availability_slot SET status = 'reserved'  
WHERE id = :slot_id;  
COMMIT;
```

7. Architecture, Security and Recovery

Four access channels

- **Patient portal:** self-service booking and history
- **Front desk:** registration, booking, check-in
- **Clinical:** queue, history, visit records
- **Management:** reports, accounts, activity review

Security controls

- HTTPS in transit; encryption at rest
- Strong password hashes + MFA for privileged roles
- Role-based, minimum-necessary access
- 15-minute staff inactivity timeout
- Append-only activity log
- Daily encrypted backups + quarterly restoration

Privacy wording

The design follows the Nigeria Data Protection Act 2023 and professional confidentiality principles. It is a classroom system proposal, not a claim of legal certification.

8. Implementation and Testing

Phases: confirm requirements → prototype → build → test → prepare data → train → pilot → deploy → review at 22, 30 and 90 days.

Critical tests: simultaneous slot booking; role bypass; invalid registration; duplicate detection; reminder retry; report reconciliation; 25-user performance; backup restoration within four hours.

Downtime: printed next-day appointment list + numbered downtime sheets; one-person back-capture and second-person verification after recovery.

9. Answer Formula

Define → Apply → Justify → Example.

"Cardinality states how many instances may relate. In Vital Spring, one patient can have many appointments, but each appointment has exactly one patient. This models repeat visits while preventing an unidentified appointment."

Closing answer

Vital Spring is technically, economically and operationally feasible. It replaces disconnected paper processes with a secure, auditable workflow that prevents scheduling conflicts, tracks live patient flow, reduces no-shows and gives management reliable attendance and waiting-time information.